Botryomycosis — Comprehensive Disease Characteristics Report

Disease: Botryomycosis **MONDO ID:** MONDO:0400006 · **MeSH:** D001857 · **OMIM:** Not applicable (non-genetic) · **Category:** Infectious Disease (bacterial pseudomycosis) **Evidence base:** Individual case reports and small case series/reviews only (no registries, cohorts, or clinical trials). All quantitative claims are qualitative/indirect.

Summary

Botryomycosis is a rare, chronic, suppurative and granulomatous BACTERIAL infection — despite the misleading "-mycosis" suffix, it is not a fungal disease. It is caused by common, non-filamentous bacteria — most frequently *Staphylococcus aureus* (NCBITaxon:1280), followed by *Pseudomonas aeruginosa* (NCBITaxon:287), with other organisms (*Escherichia coli*, *Proteus*, *Actinobacillus actinomycetemcomitans*) occasionally implicated. The defining feature is that these ordinary bacteria aggregate within tissue into compact colonies ("grains" or "granules") coated by an eosinophilic, amorphous, radiating matrix known as the **Splendore-Hoeppli phenomenon**. Clinically it takes two forms: a common **cutaneous** form (nodules, abscesses, sinuses, fistulae, ulcers, often with deep tissue and bone invasion) and a rare **visceral** form that predominantly affects the lungs and characteristically mimics carcinoma.

Botryomycosis is an **acquired, sporadic, non-genetic** condition — there is no Mendelian inheritance, no causal human gene, and no defined incidence or prevalence. It is best understood as an *aberrant host response* to a common bacterial infection rather than as disease caused by a distinct or exotic pathogen. Its central predisposing factor is **impaired cell-mediated immunity** — HIV/AIDS is the most frequently cited association — with additional risk from immunosuppressive drugs, diabetes mellitus, alcoholism, malnutrition, cystic fibrosis, and local trauma or foreign bodies. The same disease occurs naturally in animals (horses, cattle, swine), and indeed the veterinary and experimental (guinea pig) literature shaped the historical understanding of the entity.

The principal clinical importance of botryomycosis is twofold: (1) it is a **great diagnostic mimic** — of malignancy (lung cancer, sarcoma, liposarcoma, sinonasal tumor), of mycetoma, and of actinomycosis — which has led to unnecessary resections; and (2) it is **curable**. Diagnosis requires tissue biopsy (showing non-filamentous bacterial grains with Splendore-Hoeppli phenomenon) **plus** culture to identify the organism. Treatment combines **prolonged, culture-directed antibiotics** (weeks to many months) with **surgical debridement/excision**, plus restoration of immune function (e.g., antiretroviral therapy) when HIV is the driver. Localized cutaneous disease generally carries a good prognosis, but recurrence is common and disseminated/visceral disease in uncontrolled immunodeficiency can be fatal.

Section-by-Section Report

1. Disease Information

Botryomycosis is a chronic granulomatous bacterial infection of skin or (less often) viscera, characterized histologically by aggregated bacterial grains. Key identifiers: **MONDO:0400006**, **MeSH D001857**; there is **no OMIM entry** (it is not a genetic disorder), and it is not a classical Orphanet rare *genetic* disease though it is a rare condition. Common synonyms and historical/related terms include **bacterial pseudomycosis**, **granular bacteriosis**, **cutaneous granular bacteriosis**, "**bacterial ball**" (used for the sinonasal form), and — for animal disease — "botryomycoma" and (for the equine spermatic cord lesion) "scirrhous cord." The name derives from Greek *botryo-* (bunch of grapes) reflecting the grape-cluster appearance of grains. The disease was first described by Bollinger (1870) in horse lung and named by Rivolta.

Information about botryomycosis is derived almost entirely from **aggregated disease-level resources** — chiefly case reports, small case series, and narrative reviews — rather than from large EHR cohorts or registries. Only ~200–300 cutaneous cases and fewer than ~20 primary pulmonary cases have been reported historically (PMID: 37787361; PMID: 11844127).

> "The most frequent isolated etiological agents are *Staphylococcus aureus* followed by *Pseudomonas aeruginosa*. The differential diagnosis includes mycetoma, actinomycosis, and carcinomas/sarcomas." — PMID: 41128794

2. Etiology

Causal factors are infectious and mechanistic, not genetic. The primary cause is inoculation of common, non-filamentous bacteria into tissue, most often *Staphylococcus aureus* (including MRSA — PMID: 33061472), then *Pseudomonas aeruginosa*, with *E. coli*, *Proteus* spp., and *Actinobacillus actinomycetemcomitans* also reported (PMID: 17500377).

Risk factors are acquired/environmental and immunologic: - *Genetic risk factors*: **None identified.** There are no causal variants, susceptibility loci, or modifier genes for botryomycosis. This is a definitive negative finding. - *Environmental/host risk factors*: local **trauma** or surgery (inoculation route), **HIV/AIDS** and other defects of cell-mediated immunity, **immunosuppressive therapy** (e.g., methotrexate — PMID: 28034470), **diabetes mellitus**, **alcoholism**, **malnutrition**, **cystic fibrosis**, and foreign bodies. Apparent **male predominance** and occurrence at **all ages** (including children — PMID: 32173890; PMID: 11001122).

Protective factors: No genetic protective variants are described. The most relevant "protective"/curative environmental factor is **restoration of immune competence** — antiretroviral therapy produced remission in HIV-associated disease when antibiotics alone failed (PMID: 28562771).

Gene–environment interactions: Not applicable in a classical GxE sense. The operative interaction is **host immune status × bacterial infectivity**: "both the host's immune status and microbial infectivity have been implicated in the pathogenesis" (PMID: 18177488).

> "It has long been associated with defects of cellular immunity." — PMID: 28562771

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset / Course	Frequency
Chronic skin nodules	Physical manifestation	HP:0031804 (nodular skin lesion)	Subacute–chronic (weeks–months), progressive	Very common (cutaneous form)
Abscess	Clinical sign	HP:0100829 (abscess)	Chronic, relapsing	Common
Draining sinus tracts / fistulae	Clinical sign	HP:0030757 (fistula)	Chronic	Common
Skin ulcer	Physical manifestation	HP:0200042 (skin ulcer)	Chronic	Common
Grain/granule discharge ("sulfur granules"; rarely red)	Laboratory/ pathologic sign	—	Ongoing	Characteristic
Osteomyelitis / bone invasion	Clinical sign	HP:0002754 (osteomyelitis)	Late, in neglected disease	Occasional (deep cutaneous)
Pulmonary mass / cough	Clinical sign	HP:0002090 (pneumonia) / mass mimic	Chronic	Visceral form (rare)

Phenotype characteristics: **onset** is typically insidious over weeks to months; **severity** ranges from mild localized lesions to severe destructive/disfiguring disease; **progression** is chronic and relapsing. Deep cutaneous disease can invade bone: a progressive 8-year foot swelling showed radiographic/MRI bone invasion (PMID: 30429406). Grains are usually white-yellow; a unique red-grain *S. aureus* case is reported (PMID: 19736413).

Quality of life impact: No formal EQ-5D/SF-36/PROMIS data exist. Impact is inferred from disfigurement, chronic drainage, pain, and functional impairment of affected limbs; scalp/facial mutilating tumors and chronic relapsing lesions are documented (PMID: 19094129).

> "Patients with botryomycosis typically present multiple nodules, sinuses, abscesses, fistulae, and ulcers, which develop during weeks or months." — PMID: 41128794

4. Genetic / Molecular Information

Not applicable — botryomycosis has no genetic etiology. There are: - **No causal genes** (no gene symbols, no OMIM IDs). - **No pathogenic germline or somatic variants**, no ACMG classifications, no allele frequencies (gnomAD/ClinVar not relevant). - **No modifier genes, epigenetic signatures, or chromosomal abnormalities** described for the disease itself.

The only "molecular" actors are **bacterial**: *S. aureus* virulence/biofilm factors and a possible protective **glycocalyx/biofilm** around grains. This section is definitively negative for host genetics and should be recorded as such in the knowledge base.

5. Environmental Information

- **Environmental factors:** local trauma, surgical wounds, and foreign bodies provide the portal of entry. No chemical toxin, radiation, or pollution etiology.
- **Lifestyle factors:** alcoholism and malnutrition (as immunocompromising states) are contributory; there is no specific smoking/diet association beyond general immune impairment.
- **Infectious agents (central to etiology):** *Staphylococcus aureus* (**NCBITaxon:1280**; principal agent), *Pseudomonas aeruginosa* (**NCBITaxon:287**; second most common, and the likely agent in a fatal disseminated case — PMID: 13845294 — and in sinonasal "bacterial ball" — PMID: 15456177), plus *E. coli*, *Proteus*, and *Actinobacillus actinomycetemcomitans*. Co-infection is possible: *S. aureus* with *Pneumocystis carinii* in AIDS (PMID: 18177488).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Bacterial inoculation** — common non-filamentous bacteria (usually *S. aureus*), typically at a site of trauma/surgery, are introduced into tissue → *leads to*
2. **Aberrant host clearance** — in a host with impaired cell-mediated immunity (and/or a low-virulence/low-inoculum balance), the immune response neither eradicates the organism nor forms an ordinary resolving abscess → *results in*

3. **Bacterial aggregation into grains** — organisms clump into compact colonies (grains/granules) → *leads to*
4. **Splendore-Hoeppli coating** (*inferred*) — grains become surrounded by an eosinophilic amorphous matrix of host immunoglobulin, complement, and degranulated eosinophil/neutrophil products, often with radiating "clubs" → *which*
5. **Immune/antibiotic shielding** (*inferred*) — this matrix (± bacterial glycocalyx/biofilm) shields organisms from phagocytosis and antibiotic penetration → *leads to*
6. **Chronic pyogranulomatous inflammation** — persistent influx of neutrophils, macrophages, multinucleated giant cells, and plasma cells with fibrosis → *results in*
7. **Clinical manifestation** — chronic nodules, draining sinus tracts/fistulae, ulcers, and local tissue/bone destruction (or a visceral pulmonary mass).

Where the chain **branches**: with adequate host immunity + eradication → ordinary abscess/healing (no botryomycosis); with continued immune failure → dissemination and potentially fatal visceral disease.

Categories mapped onto the chain: - *Cellular processes:* inflammation (**GO:0006954**, inflammatory response), phagocytosis (**GO:0006909**), humoral immunity (**GO:0002455**). - *Immune involvement:* impaired cell-mediated immunity is upstream/permissive; chronic granulomatous and humoral responses are downstream. - *Protein/biochemical:* Splendore-Hoeppli matrix = host immunoglobulin + complement + eosinophil major basic protein; possible bacterial biofilm glycocalyx. - *Tissue damage:* suppuration, fibrosis, and osteolysis (bone invasion) via chronic pyogranulomatous destruction. - *Cell types (CL):* neutrophil (**CL:0000775**), macrophage (**CL:0000235**), eosinophil (**CL:0000771**), plus multinucleated giant cells and plasma cells.

Mechanistic caveats are explicit in the literature: botryomycosis is "an atypical reaction of the host to a common bacterial infection" (PMID: 9435048) and "an unusual response to infection with common bacteria" (PMID: 11243368), but "the pathobiology remains unknown" (PMID: 23743320) — steps 4–5 are **inferred**, not fully demonstrated.

> "both the host's immune status and microbial infectivity have been implicated in the pathogenesis of this unusual clinical condition" — PMID: 18177488

7. Anatomical Structures Affected

- **Organ level (primary):** skin (**UBERON:0002097**) in the cutaneous form; lung (**UBERON:0002048**) in the visceral form. **Secondary:** bone (**UBERON:0001474**; osteomyelitis/invasion), subcutaneous soft tissue, and mucosal sites (sinonasal tract, tongue — PMID: 5235476, lacrimal sac — PMID: 29039982).
- **Body systems:** integumentary (primary), respiratory (visceral), musculoskeletal (bone invasion), and reticuloendothelial/immune.
- **Tissue/cell level:** dermis and subcutis; pyogranulomatous infiltrate of neutrophils (**CL:0000775**), macrophages (**CL:0000235**), eosinophils (**CL:0000771**), giant cells, plasma cells.
- **Subcellular:** no specific human organelle involvement; relevant compartments are the bacterial cell wall and the extracellular Splendore-Hoeppli matrix (GO extracellular region).
- **Localization/lateralization:** cutaneous lesions favor **extremities** (foot/hand), **head/neck**, and unusual sites; typically **unilateral/localized** at the inoculation site, though disseminated bilateral disease occurs in immunodeficiency.

8. Temporal Development

- **Onset:** any age; **insidious/chronic** onset over weeks to months, usually after trauma.
- **Progression:** slowly progressive and **relapsing**; neglected cutaneous disease progresses over years (e.g., 8-year foot course — PMID: 30429406) to deep/bone destruction; visceral disease can disseminate.
- **Course pattern:** chronic, relapsing-remitting with therapy; **recurrence is common** after treatment.
- **Duration:** chronic if untreated; potentially curable with prolonged combined therapy.
- **Critical period:** early diagnosis (before bone/visceral involvement and before unnecessary oncologic surgery) is the key window for cure.

9. Inheritance and Population

- **Epidemiology:** rare; **no defined incidence or prevalence**. ~200–300 cutaneous and <20 primary pulmonary cases historically reported; one review compiled 18 primary pulmonary cases (PMID: 37787361); another was the 16th pulmonary case (PMID: 11844127).
- **Inheritance: None** — not genetic; no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency applies.

- **Demographics:** worldwide, no defined endemic regions; **apparent male predominance**; **all ages** including children; risk clusters in the immunocompromised. A recent Mexican case series adds regional data (PMID: 42220824).

> "Botryomycosis occurs more commonly among immunocompromised patients, although some cases have also been reported in immunocompetent patients." — PMID: 32173890

10. Diagnostics

Diagnosis rests on histopathology + culture. - **Biopsy/histopathology:** suppurative/pyogranulomatous inflammation with **bacterial grains** surrounded by an **eosinophilic Splendore-Hoeppli** matrix and radiating "clubs"; grains contain Gram-positive cocci or Gram-negative bacilli and are **PAS/Grocott positive but lack filaments** (distinguishing from actinomycosis/mycetoma) (PMID: 32173890; PMID: 28034470; PMID: 15456177). - **Microbiology/culture:** essential to identify the organism and guide therapy. - **Imaging:** X-ray/MRI for bone invasion; CT for pulmonary/sinonasal masses (often reveals a mass mimicking neoplasm with osseous erosion). - **Genetic/omics testing: not applicable.** - **Clinical criteria:** no formal DSM/ICD diagnostic criteria; diagnosis is pathologic. - **Differential diagnosis: mycetoma** (grains contain filaments; fungal or actinomycetoma), **actinomycosis** (filamentous Gram-positive), and **malignancy** (lung cancer, sarcoma, liposarcoma, sinonasal tumor). Note histologic overlap with fungal infections showing Splendore-Hoeppli (e.g., basidiobolomycosis — PMID: 42266393), reinforcing the need for special stains and culture.

> "The diagnosis of botryomycosis is based on culture of the granules revealing gram-positive cocci or gram-negative bacilli." — PMID: 23743320
>
> "In immunocompromised patients, botryomycosis can resemble fungal infection both clinically and histologically; therefore, definitive diagnosis requires tissue sampling along with histological and microbiological analysis." — PMID: 38036751

11. Outcome / Prognosis

- **Survival/mortality:** generally **good** for localized cutaneous disease with combined therapy; **fatal outcomes** occur with disseminated visceral disease and uncontrolled immunodeficiency (fatal AIDS-associated case — PMID: 38036751; fatal disseminated

visceral case probably due to *P. aeruginosa* — PMID: 13845294). No formal survival statistics exist.

- **Morbidity/function:** disfigurement, chronic drainage, limb dysfunction, bone destruction.
- **Complications:** osteomyelitis, sepsis/bacteremia with pulmonary abscesses (PMID: 28034470), dissemination.
- **Recovery:** cure achievable; **recurrence is common** and long therapy is required.
- **Prognostic factors:** host immune status (dominant), lesion extent/depth (bone/visceral), organism, and timeliness of diagnosis. No molecular prognostic biomarkers.

12. Treatment

Principle: prolonged, culture-directed antibiotics + surgical debridement/excision, plus immune restoration when relevant.

Regimen / intervention	Setting	Outcome	Source
Excision + oral TMP-SMX (240 mg b.i.d.) × 2 wk	Immunocompetent child, cutaneous	No recurrence	PMID: 32173890
Excision + penicillin G → clindamycin + cotrimoxazole × 18 mo	Scalp granular bacteriosis	Disease-free at 4 y	PMID: 19094129
Cefazolin	Red-grain *S. aureus* foot	Responded well	PMID: 19736413
Fluoroquinolone (vs *A. actinomycetemcomitans*)	Hand, dual infection	Resolved only after fluoroquinolone	PMID: 17500377
Cloxacillin-gentamicin → rifampicin-ofloxacin × 2 mo	*S. aureus* sepsis + pulmonary abscesses	Managed bacteremia	PMID: 28034470
Antiretroviral therapy (darunavir + raltegravir)	HIV facial pyoderma	Remission at 5 y after ART	PMID: 28562771
Complete surgical evacuation	Sinonasal "bacterial ball"	Curative	PMID: 15456177

- **Pharmacotherapy classes / NCIT anchors:** TMP-SMX (**NCIT:C1055**), penicillins, clindamycin, cephalosporins (cefazolin), fluoroquinolones (ofloxacin), rifampicin, aminoglycosides (gentamicin) — chosen by culture/sensitivity.

- **Surgical:** debridement/excision (**NCIT:C51696**, debridement); complete evacuation can be curative for localized visceral/mucosal disease.
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno-oncology): Not applicable.**
- **Pharmacogenomics:** not applicable.
- **Strategy:** combine surgery + long antibiotics; **always assess/treat immune status** (ART in HIV). Recurrence-prone → extended courses.

> "Surgical debridement is the treatment of choice in such cases with an assessment of the immune status. Long term antibiotic treatment is required in all conditions as recurrence is common." — PMID: 29039982
>
> "The patient was successfully treated by excision and oral trimethoprim/sulfamethoxazole (240 mg b.i.d.) for 2 weeks as adjuvant therapy. No recurrence was noted following treatment." — PMID: 32173890

13. Prevention

- **Primary prevention:** good **wound care** and prompt treatment of skin infections; management of predisposing conditions (control HIV with ART, control diabetes, address malnutrition/alcoholism).
- **Secondary prevention:** early biopsy of chronic non-healing nodules/sinuses to diagnose before deep/visceral spread and to avoid unnecessary oncologic surgery.
- **Tertiary prevention:** prolonged antibiotics + surgery to prevent recurrence, bone destruction, and dissemination; immune restoration.
- **Immunization / genetic screening / counseling: Not applicable** (no vaccine; not heritable).
- **Public health:** general infection-control and immunosuppression management; no vector control needed.

14. Other Species / Natural Disease

Botryomycosis is a genuine **cross-species (shared-agent) disease** — the same bacteria and the same host-response pathology occur in animals and humans, though it is not classically transmitted person-to-person or animal-to-person. - **Taxonomy of hosts:** horse (*Equus caballus*), cattle (*Bos taurus*), swine (*Sus scrofa*), guinea pig (experimental), and human (*Homo sapiens*). - **Natural disease:** most common as equine **wound infection** (castration site /

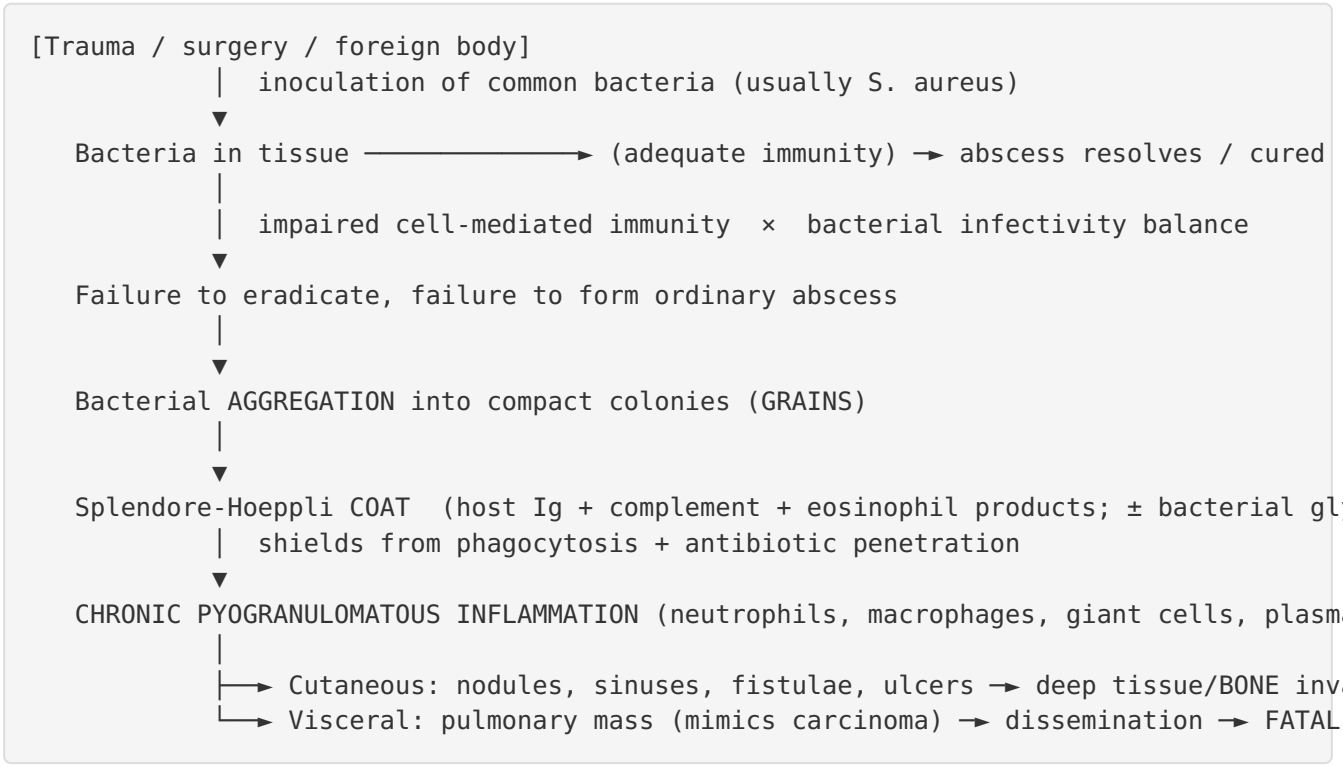
"scirrhous cord," spermatic-cord granuloma) and as **mastitis** in cows and sows; rare pulmonary disease (documented in a Scottish Highland steer, *S. aureus* cultured — PMID: 11243368). - **Comparative pathology:** histology in animals **matches** human disease — pyogranulomatous inflammation with eosinophilic grains and radiating clubs around Gram-positive cocci — indicating a **conserved host-response mechanism** across mammals. - **Orthologous genes:** not applicable (no host causal gene).

> "Among domestic species, staphylococcal botryomycosis is most common as a wound infection in horses or as mastitis in cows and sows." — PMID: 11243368

15. Model Organisms

- **Genetic models: None** — because the disease has no genetic basis, there are no knockout/knock-in/transgenic models.
- **Induced/experimental models:** historically, **guinea pigs** were used to study experimental botryomycosis (pulmonary botryomycosis noted "rare in horses, humans, and guinea pigs" — PMID: 11243368); naturally occurring disease in **horses, cattle, swine** serves as the comparative/large-animal model.
- **Applications:** these models illuminate grain formation, the Splendore-Hoeppli response, and the host-immunity/bacterial-infectivity balance.
- **Limitations:** no standardized modern model organism system; mechanistic detail (steps 4–5 of the causal chain) remains under-characterized.

Mechanistic Model / Interpretation



The unifying interpretation is that **botryomycosis is not a distinct pathogen's disease but an aberrant, grain-forming host response to ordinary bacteria**, gated by host immunity. This explains its three defining clinical truths: it *mimics* malignancy/mycetoma/actinomycosis (because grains + granulomatous mass resemble tumors and other grain diseases); it is *curable* (because the organisms are common and antibiotic-susceptible); and it is *recurrence-prone and dangerous in immunodeficiency* (because the permissive lesion is the host's failed clearance, best fixed by restoring immunity alongside antibiotics and surgery).

Evidence Base

PMID	Title (abbrev.)	Role in report
41128794	*Cutaneous botryomycosis: a comprehensive review*	Etiology (S. aureus > P. aeruginosa), lesion morphology, treatment, differential dx
37787361	*Primary pulmonary botryomycosis*	Two forms; visceral rarity; pulmonary predominance
32173890	*Cutaneous botryomycosis in an 8-yo immunocompetent boy*	Splendore-Hoeppli histology; occurrence in immunocompetent; TMP-SMX cure
38036751	*Botryomycosis mimicking sarcoma / AIDS presentation*	Immunocompromised predominance; need for tissue + microbiology; fatal outcome
28562771	*Facial botryomycosis-like pyoderma in HIV*	Cellular immunity link; ART-induced remission
18177488	*S. aureus + P. carinii in AIDS*	Dual host-immune × microbial pathogenesis
9435048	*Botryomycosis mimicking a liposarcoma*	"Atypical host reaction" framing; cancer mimicry
23743320	*Pulmonary botryomycosis mimicking lung cancer*	Culture-based diagnosis; "pathobiology remains unknown"
11844127	*Primary pulmonary botryomycosis vs lung cancer*	Curable differential of lung cancer; rarity
11243368	*Pulmonary botryomycosis in a Scottish Highland steer*	Animal disease; conserved pyogranulomatous grain histology
30429406	*Cutaneous botryomycosis of the foot*	Deep tissue/bone invasion; destruction/ disfigurement
15456177	*Sinonasal "bacterial ball" (P. aeruginosa)*	Mucosal disease; neoplasm mimic; surgical cure
17500377	*A. actinomycetemcomitans botryomycosis*	Non-staph etiology; fluoroquinolone cure
28034470	*Cutaneous granular bacteriosis in S. aureus septicaemia*	Splendore-Hoeppli, immunosuppression (MTX), dual antibiotics
19094129	*Granular bacteriosis of head/neck*	18-month triple antibiotics + surgery; 4-year cure

PMID	Title (abbrev.)	Role in report
29039982	*Botryomycosis of the lacrimal sac*	Surgery + long antibiotics; immune assessment; recurrence
19736413	*Red-grain botryomycosis (S. aureus)*	Grain color variant; cefazolin response
13845294	*Disseminated visceral botryomycosis (fatal)*	Fatal P. aeruginosa dissemination
12602981	*Botryomycosis in HIV*	AIDS ulcers; excision + antimicrobials
11001122	*Botryomycosis in a child with AIDS*	Pediatric immunocompromised case
42220824	*Case series, Mexico*	Regional case series; non-filamentous organisms
33061472	*MRSA botryomycosis*	MRSA as agent
5235476	*Botryomycosis of the tongue*	Mucosal (oral) site

Papers on gastrointestinal basidiobolomycosis (PMID: 42266393, 42471849, 41634777, 42776653) are **not botryomycosis** but were reviewed because they share the Splendore-Hoeppli phenomenon and the "mimics malignancy/IBD" behavior — they usefully delimit botryomycosis from its fungal histologic look-alikes and underscore that Splendore-Hoeppli is non-specific.

Limitations and Knowledge Gaps

1. **Evidence quality is low-tier:** the entire knowledge base is case reports/series and narrative reviews — no cohorts, no trials, no registries. All frequencies are qualitative.
2. **No epidemiologic denominators:** true incidence/prevalence, sex ratio, and geographic distribution are unknown.
3. **Mechanism partly inferred:** steps 4–5 (Splendore-Hoeppli shielding, biofilm/glycocalyx role) are hypotheses; "the pathobiology remains unknown" (PMID: 23743320). The molecular determinants of why the *same* bacteria cause a grain-forming granuloma in one host and a simple abscess in another are undefined.

4. **No standardized treatment protocol:** antibiotic choice, duration, and the surgery-vs-medical-therapy balance are empirical and heterogeneous.
5. **No molecular/omics data** and no modern model organism system exist for this disease.

Proposed Follow-up Experiments / Actions

1. **Systematic pooled analysis** of published cases (extract organism, host immune status, site, therapy, duration, recurrence, outcome) to derive quantitative treatment-response and recurrence rates.
2. **Mechanistic studies of grain formation:** compare biofilm/glycocalyx and Splendore-Hoeppli matrix composition (host Ig/complement/eosinophil MBP) between botryomycosis grains and ordinary abscesses; test whether cell-mediated immune deficiency is *necessary* using immunodeficient vs immunocompetent animal inoculation models.
3. **Prospective diagnostic pathway** in chronic non-healing cutaneous nodules/pulmonary masses: mandate biopsy + culture + special stains (Gram/PAS/Grocott) before oncologic resection, to quantify avoided surgeries.
4. **Molecular characterization** of *S. aureus*/*P. aeruginosa* isolates (virulence/biofilm genotypes) from grains vs matched abscesses to test the "microbial infectivity" arm of the causal model.
5. **Immune-restoration studies:** formally evaluate adjunctive ART/immune optimization outcomes in HIV-associated botryomycosis versus antibiotics/surgery alone.

Ontology anchors used in this report: MONDO:0400006; MeSH D001857; NCBITaxon:1280 (*S. aureus*), NCBITaxon:287 (*P. aeruginosa*); UBERON:0002097 (skin), UBERON:0002048 (lung), UBERON:0001474 (bone); HP:0100829 (abscess), HP:0030757 (fistula), HP:0200042 (skin ulcer), HP:0002754 (osteomyelitis); GO:0006954 (inflammatory response), GO:0006909 (phagocytosis), GO:0002455 (humoral immune response); CL:0000775 (neutrophil), CL:0000235 (macrophage), CL:0000771 (eosinophil); NCIT:C1055 (TMP-SMX), NCIT:C51696 (debridement).
